2 Determination of enthalpy change of neutralisation

You will determine the concentration of sulfuric acid by reacting with a known concentration of sodium hydroxide using a thermometric method. You are to also determine the enthalpy change of neutralisation per mole of water formed when these solutions react. The equation for the reaction is shown.

$$2\ NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2\ H_2O(l)$$

FA 2 is dilute sulfuric acid, H_2SO_4.

FA 5 is 1.90 mol dm^{-3} sodium hydroxide, NaOH.

In this question, you will carry out a series of experiments where different volumes of **FA 5** and **FA 2** are mixed together.

You will determine the temperature change of the mixture, ΔT, of each experiment and then analyse your results graphically in order to determine the

- concentration of H_2SO_4 in **FA 2**
- maximum temperature change, ΔT_{max}
- value for the enthalpy change of neutralisation, ΔH_{neut}

(a) Procedure

1. Place a polystyrene cup inside a second polystyrene cup and place both cups in a glass beaker. The retort clamp provided may be used to clamp the beaker to prevent it from tipping.
2. Use a 50 cm^3 measuring cylinder to transfer 50.0 cm^3 of **FA 5** into the polystyrene cup.
3. Place the lid with a hole in the centre on the cup and insert the thermometer through the lid. Stir the **FA 5** solution gently with the thermometer. Read and record the initial temperature of the solution of **FA 5** as $T_{initial}$.
4. Place 10.0 cm^3 of **FA 2** into another 50 cm^3 measuring cylinder.
5. Transfer the **FA 2** from the measuring cylinder into the polystyrene cup and close the lid. Stir the mixture gently with the thermometer.
6. Read and record the maximum temperature of the mixture, T_{max}, and the volume of **FA 2** added.
7. Rinse and dry the polystyrene cup and the thermometer with paper towel.

8. Repeat steps **1** to **6** using **5** different volumes of **FA 5,** mixing it with the appropriate volumes of **FA 2** to be used in each experiment such that the total volume of the reaction mixture is **60.0 cm^3**.
9. Calculate ΔT for each experiment.

In an appropriate format in the space provided, prepare a table in which to record for each experiment

- all volumes, $V_{FA\,2}$ and $V_{FA\,5}$, to an appropriate level of precision
- all values of temperature, $T_{initial}$, T_{max} and ΔT to an appropriate level of precision.

Results

M16

M17

M18

M19

[4]

(b) Plot a graph of ΔT against $V_{FA\,2}$ added on the grid in Fig. 2.1. Your scale on the y-axis should allow for extrapolation above the highest temperature change recorded. Label any points you consider to be anomalous.

Draw two lines of best fit, one for the rise in temperature and one for after the maximum temperature has been reached. Extrapolate (extend) both lines until they intersect.

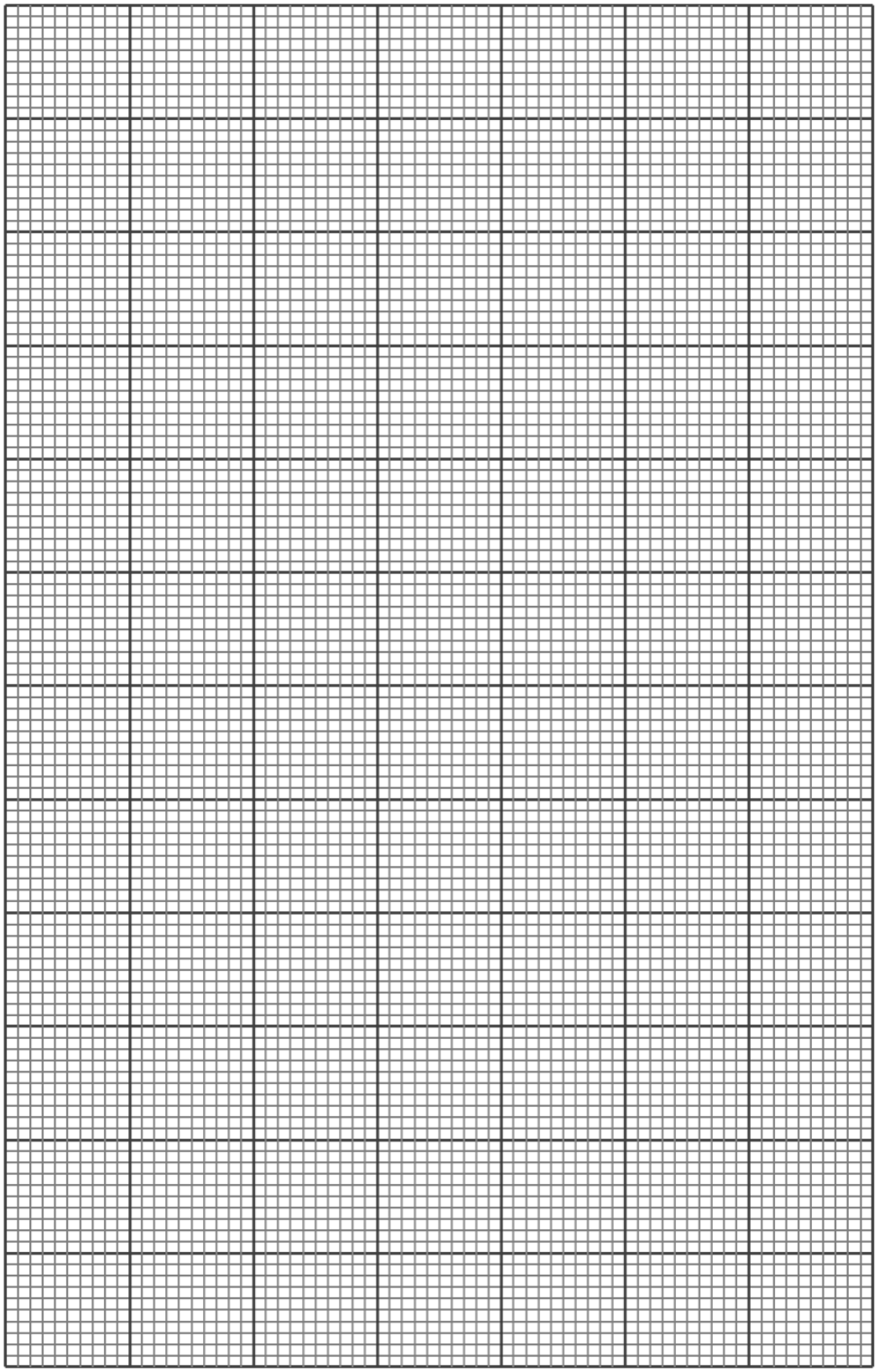

Fig. 2.1

[3]

M20

M21

M22

2 **(c)** **(i)** From Fig. 2.1, determine the maximum temperature change, ΔT_{max}, and the V_{FA2} required to achieve this maximum temperature change.

Maximum temperature change, ΔT_{max} : ..... °C ☐ M23

Volume of **FA 2**, V_{FA2} : ..... cm^3 **[2]** ☐ M24

(ii) Calculate the number of moles of **FA 5** reacted and hence, calculate the concentration of **FA 2**.

Number of moles of **FA 5** reacted: ..... ☐ M25

Concentration of **FA2**: ..... mol dm^{-3} **[2]** ☐ M26

(iii) Calculate the heat change, q, at the point of neutralisation, in your experiment using the ΔT_{max} value you deduced in **(c)(i)**.

You should assume that the specific heat capacity of the solution is 4.18 J g^{-1} K^{-1}, and that the density of the solution is 1.00 g cm^{-3}.

q: ..... **[1]** ☐ M27

2 **(c)** **(iv)** Hence, calculate the enthalpy change of neutralisation, ΔH_{neut}, per mole of water formed.

The equation for the reaction is shown.

$2\ NaOH(aq) + H_2SO_4(aq) \rightarrow Na_2SO_4(aq) + 2\ H_2O(l)$

Include the sign of ΔH_{neut} in your answer.

ΔH_{neut} = ……..

[2]

M28

M29

(d) A student repeated the same experiment but used CH_3COOH instead of H_2SO_4 of the same concentration. Suggest and explain how the maximum temperature change will be affected.

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…… **[2]**

M30

M31

[Total: 16]